BSC-MD: Developing an Android App

Proposal Sheet

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Idea:

My idea is to develop a ticket booking app for a cinema. Technically this project will allow me to integrate knowledge and skills that I have learned from previous modules including Object-Oriented Programming, Software Development, System Design, and Concurrent Development. In Concurrent Development module I worked an example which is about seat booking to utilize multi-threading mechanism. Which gave me the idea of building a real APP in this mobile development module.

The main functions of the application will include browsing the movies list, selecting the date and time, booking seats, and generating QR code for the ticket. Seats booing process will be applied in a thread-safe booking mechanism to prevent the same seat from being booked at the same time by different booking tasks.

In addition, the APP will use GPS to determine the user's location and direct to Google Map through the cinema intent.

Planned Activities/Layouts:

There will have several activities. The main page will display the basic cinema information, some movie posters, location and home & my tickets buttons.

A movie page will show all the current movies with description and let users to select date and showtime.

A seat page will allow users to choose available seats and confirm booking.

The application will also include a ticket page displaying the booking details and QR code, and my ticket page for viewing booked tickets.

Sensor use:

GPS / Device Location

The application will use user's current location to calculate the distance between the device and the cinema. Then the user will be able to open the cinema location by Google Map.

Storage:

The APP will use local files for persistent storage to save information such as movies, showtimes and booing records. This will allow booked tickets to remain available after the application is closed and reopened.